AI Unboxed and Jobs: A Novel Measure and Firm-Level Evidence from Three Countries*

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Abstract

Despite widespread concern, the labour-market impacts of artificial intelligence (AI) remain hard to quantify. We construct a Dynamic AI Occupational Exposure (DAIOE) index—tracking annual frontier gains across nine AI subdomains (2010–2023)—and link it to employer–employee registers in Denmark, Portugal, and Sweden. While AI exposure has no net effect on total firm employment, it is consistently tied to compositional up-skilling: firms with higher DAIOE scores increase their high-to-low skill employment ratios. Firms with higher DAIOE scores tend to expand high-skill white-collar positions and reduce low-skill clerical roles. Blue-collar effects average out but vary sharply by subdomain. Our findings underscore the need to unpack "AI" into its component technologies when evaluating labour-market outcomes.

Keywords: Artificial intelligence; Labour demand; Multi-country firm-level evidence

JEL Codes: E24, J23, J24, N34, O33.

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1. Introduction

Recent breakthroughs in artificial intelligence (AI)—from deep-learning advances in vision and language to the rise of generative models—have galvanised public debate about labour-market impacts (e.g., OECD, 2023, Korinek, 2023, Hudson, 2024, Harwell, 2025, Trivedi, 2025). At its core, AI comprises methods and software enabling agents to perceive their environment, learn, and act to achieve specific objectives (Russel and Norvig, 2010). Contemporary systems perform tasks once thought to require human intelligence—processing vast datasets under varying degrees of supervision, assisting decision-makers, generating content, and even making autonomous decisions (OECD, 2024). This recent "AI summer", driven by rapid advances at the technological frontier, has significant implications for work, particularly in white-collar jobs.

Conceptually, AI may both substitute for and complement human labour.[1] Early work presumed that AI was yet another automation technology, displacing workers in routine tasks (e.g., Acemoglu and Restrepo, 2019, Aghion *et al.*, 2019a). More recent studies, however, emphasise AI's potential to complement labour—boosting innovation, new product introductions, and thus raising demand in skill-intensive occupations (e.g., Babina *et al*, 2024, Bessen *et al*, 2022, Hirvonen *et al.*, 2025, Handa *et al.*, 2025).

Yet, assessing AI's real-world labour-market effects remains elusive: AI spans diverse, hard-to-measure methods—from large-scale language models to vision systems and game-playing agents—and workers face uneven exposure.[2] To address this gap, we (i) develop and validate a Dynamic AI Occupational Exposure (DAIOE) measure—disaggregated by nine subdomains (e.g., language modelling, image recognition) and year (2010–2023)—to capture how frontier advances translate into occupational exposure to both automation and complementation, while accounting for the role of social interaction in work; and (ii) link DAIOE to

[1] AI may even create entirely new tasks that humans cannot perform (Brynjolfsson, 2022).

[2] Over a dozen experiments examine generative AI's impact on narrow tasks—drafting emails, coding snippets, etc.(e.g., Brynjolfsson *et al*, 2025, Dell'Acqua *et al*, 2023, Dell'Acqua, 2024, Noy and Zhang, 2023, Otis *et al.*, 2023, Peng *et al.*, 2023).

rich employer–employee registers in Denmark, Portugal, and Sweden to trace how distinct AI advances map into compositional labour-demand shifts.

We transform 140 benchmark frontiers from the Electronic Frontier Foundation (EFF) and Papers With Code (PWC) into annual gains for each AI application, then apply the Felten *et al.* (2018) mapping matrix to convert those gains into ability-level exposure shifts. We aggregate these, via ONET ability weights, into occupation-specific, year-on-year DAIOE scores.[3] We further adjust for occupations' social-skill intensity to reflect the greater resilience of high-interaction roles. Unlike prior work, which typically produces a single, aggregated, and time-invariant exposure score—and often relies on indirect assessments of AI progress, such as human or generative AI task labeling (Felten *et al.*, 2018, 2021, Webb, 2019, Gmyrek *et al.*, 2023, Lábaj *et al.*, 2024, Prytkova *et al.*, 2024, Eloundou *et al.*, 2024)—our approach directly incorporates measured technological advancements over time.[4]

Our novel measure enables us to analyse the evolution of AI and its effects on occupational exposure in recent years. By examining the most recent data point, for 2023, we gain insight into the current state of exposure. Moreover, the measure can be adapted to simulate future scenarios. In our interpretation, exposure reflects the potential applicability of AI technologies to a given occupation; however, it does not predict whether, when, or through which channels this exposure will translate into changes in labour demand. The most AI-exposed occupations tend to be cognitive, non-physical, and low in social interaction—in other words, white-collar roles, particularly those involving limited interpersonal engagement.[5]

By merging DAIOE with longitudinal linked-employer–employee data, we trace how variation

[3]We build on the seminal work of Felten *et al.* (2018, 2021), who assess occupational exposure to AI by linking AI progress to the importance of abilities across occupations. Occupational AI exposure measures have been shown to correlate strongly with changes in establishment-level hiring patterns, underscoring their relevance for capturing the economic impact of AI developments (Acemoglu *et al.*, 2022).

[4]An alternative strategy employs natural language processing to assess textual similarity between AI-related patents and occupational descriptions or task statements (Webb, 2019, Lábaj *et al.*, 2024, Prytkova *et al.*, 2024). However, since many AI innovations are software-based and thus not easily patentable, we instead rely on benchmark datasets from AI research to measure capability progress.

[5]The measure can be combined with data on occupational structures—e.g., of firms, industries, regions, or countries—to assess overall workforce exposure to AI technologies.

in AI exposure—both overall and by subdomain—relates to firms' employment of high- and low-skill white-collar and blue-collar workers. This combined methodological and empirical approach allows us to document a clear compositional up-skilling, alongside heterogeneous labour demand patterns across occupations and countries.

We apply our analysis to data from three European Union countries—Denmark, Portugal, and Sweden—that exhibit significant differences in labour market rigidity, workforce composition, industrial structure, productivity, digital intensity, and AI adoption rates. For example, in 2024, 28 percent of enterprises in Denmark and 25 percent in Sweden utilized AI technologies, compared to lower adoption rates in all other EU countries, including Portugal (Eurostat, 2024). This cross-country heterogeneity provides a natural laboratory to explore how varying economic and structural factors influence the differential impacts of AI advancements on workers, businesses, and national economies.

Previous research has primarily examined the average effects of AI on labour markets, typically relying on aggregated or limited-sample data, most often from the United States (e.g., Georgieff and Hyee, 2022, Felten *et al.*, 2019, Fossen and Sorgner, 2022, Lane and Saint-Martin, 2021). A few studies have drawn on online job postings, again largely U.S.-based (e.g., Acemoglu *et al.*, 2022, Babina *et al*, 2023). However, identifying the detailed and heterogeneous impacts of AI is critical (Seamans and Raj, 2018, Frank *et al.*, 2019, Zolas *et al.*, 2021, OECD *et al.*, 2022, OECD, 2023, Korinek, 2023). Using our DAIOE measure, we uncover divergent associations across worker groups and AI subdomains. Indeed, recent advances in AI have been highly uneven across time and technologies: image and speech recognition progressed markedly in the early 2010s; machine translation saw key developments in the mid-2010s (Zhang *et al.*, 2021); and language modelling experienced a breakthrough around 2020. This paper is the first to systematically examine these nuanced interactions between the evolution of AI technologies and labour market outcomes.

Estimating firm-level labour-demand regressions using DAIOE linked to each firm's baseline occupational mix, we find a robust association between AI exposure and an increase in the high-to-low skill ratio—evidence of compositional up-skilling. A one-standard-deviation rise in DAIOE is associated with a 1.6 percentage point (pp) increase in the mean of firms' high-to-low skill ratio in Denmark (1.3 pp in Sweden; 3.6 pp in Portugal). Consistent with this up-skilling, higher AI exposure also correlates with a reduction in low-skill white-collar roles and a rise in high-skill white-collar positions. Subdomain estimates further suggest that advances in image and language technologies most strongly drive these shifts.

The remainder of the paper is organised as follows. Section 2. introduces our Dynamic AI Occupational Exposure (DAIOE) measure and presents descriptive statistics. Section 3. lays out the conceptual framework. Section 4. describes the three countries, the register data, and our estimation and identification strategy. In Section 5., we report the econometric results on AI exposure and firm-level labour demand and Section 6. concludes. Robustness checks and extended results appear in the Appendix, with further methodological details provided in the Online Appendix.

2. A novel measure for occupational AI exposure: DAIOE

To measure the potential applicability of AI across occupations, we integrate data on AI advancements with information about occupational work content to construct a dynamic AI occupational exposure measure (DAIOE). Structured as a panel, this measure provides unique values for each occupation-year, capturing the evolution of AI exposure over time. Covering the years 2010–2023—a period marked by significant advancements in various AI technologies—DAIOE enables analysis of temporal trends in occupational exposure.[6]

Our methodology builds upon the seminal work of Felten, Raj, and Seamans (Felten *et al.*, 2018, 2021). Specifically, Felten *et al.* (2021) introduced a forward-looking measure, the AI Occupational Exposure (AIOE), designed to assess the prospective effects of AI on occupations. In contrast, our approach adopts a retrospective perspective, akin to that of Felten

[6]Further methodological details can be found in the Online Appendix.

et al. (2018), focusing on recent data to analyse AI exposure. We incorporate the term *dynamic* to emphasise the time-variant nature of our measure, capturing the evolution of AI exposure across occupations over the period 2010–2023.

2.1. Construction of the Measure

To measure AI progress, we make use of *benchmarks* that have been used in AI research to test AI performance. Data on benchmarks are from the Electronic Frontier Foundation (EFF) and Papers With Code (PWC) on AI progress across applications or sub-domains.[7] To the best of our knowledge, these databases represent the most extensive publicly accessible sources of AI research data.[8]

Following Felten *et al.* (2018, 2021), we classify AI technology into nine main AI applications, see Table 1. These nine applications are categorised into three primary areas—games, vision, and language—and have been validated by AI researchers as representative of key AI research domains during the study period.

TABLE 1

AI Applications Included in the Exposure Measure

Area	*Application*	*Benchmark Example*	*#bm*
Games	Abstract strategy games	Chess	2
	Real-time video games	Atari games	58
Vision	Image recognition	Imagenet Image Recognition Challenge	13
	Image comprehension	COCO Visual Question Answering	9
	Image generation	Generative models of CIFAR-10 images	7
Language	Reading comprehension	bAbi 20 QA reading comprehension	19
	Language modelling	Penn Treebank	9
	Translation	En-De Translation BLEU scores	13
	Speech recognition	Automatic Speech Recognition on LRS2	10

Notes: The table lists the AI applications (subdomains) used to construct the occupational exposure measure, along with examples of benchmarks used to track technological progress in each subdomain, as well as the number of benchmarks per AI application. For a complete list of the 140 benchmarks, see the Online Appendix.

[7] Data are available at: `https://www.eff.org/ai/metrics` and `https://paperswithcode.com`. In the EFF data, the benchmarks are referred to as *metrics*.

[8] Data from the EFF have been extensively used in various research fields, such as computer science, economics and management (Felten *et al.*, 2018, 2021), and PWC is widely used within the machine learning community (Martínez-Plumed *et al.*, 2021).

Within each AI application, the EFF and PWC databases provide performance data on a range of benchmarks commonly used in AI research. For instance, in image recognition, the ImageNet Large Scale Visual Recognition Challenge (ILSVRC) has served as a prominent benchmark, where AI systems compete to accurately label images.[9] Similarly, in the domain of abstract strategy games, benchmarks include AI performance in chess. Across 140 such benchmarks, we compile longitudinal data on AI performance over time.

The benchmarks in our database exhibit diverse scales: some report performance as percentages (e.g., accuracy rates in ImageNet), while others use absolute metrics (e.g., points scored in Atari games). To facilitate comparability across these varied benchmarks, we adopt the rescaling methodology proposed by Felten *et al.* (2018), transforming each metric so that a linear increase reflects an exponential improvement in performance.[10]

We calculate a state of the art (SOTA) frontier for each benchmark, representing the highest AI performance to date. This frontier, prior to scaling, for the benchmark CIFAR-10 Image Recognition is illustrated in Figure 1.[11]

To derive an overall progress curve for each AI application, we aggregate performance across benchmarks. We estimate year-by-year progress by computing the average slope of the performance frontiers for each benchmark annually, denoted as Δp_{it} for application i at time t. Summing these yearly changes yields cumulative progress curves for all nine applications, as illustrated in Figure 2.

To connect AI advancements to occupational tasks, we utilize the Occupational Information Network (O*NET) database, sponsored by the U.S. Department of Labor (Handel, 2016). O*NET provides standardized information on occupational requirements, including 52 worker abilities that capture key individual characteristics affecting job performance, such as oral communication, reasoning, vision, and physical strength.[12]

[9] Although the competition concluded in 2017, its datasets continue to be widely utilised.
[10] This captures diminishing marginal gains in AI advancements as they approach human-level proficiency.
[11] The scaled frontiers for the full set of image recognition benchmarks are shown in the Online Appendix.
[12] For the complete list of abilities, along with their categories and descriptions, see the Online Appendix.

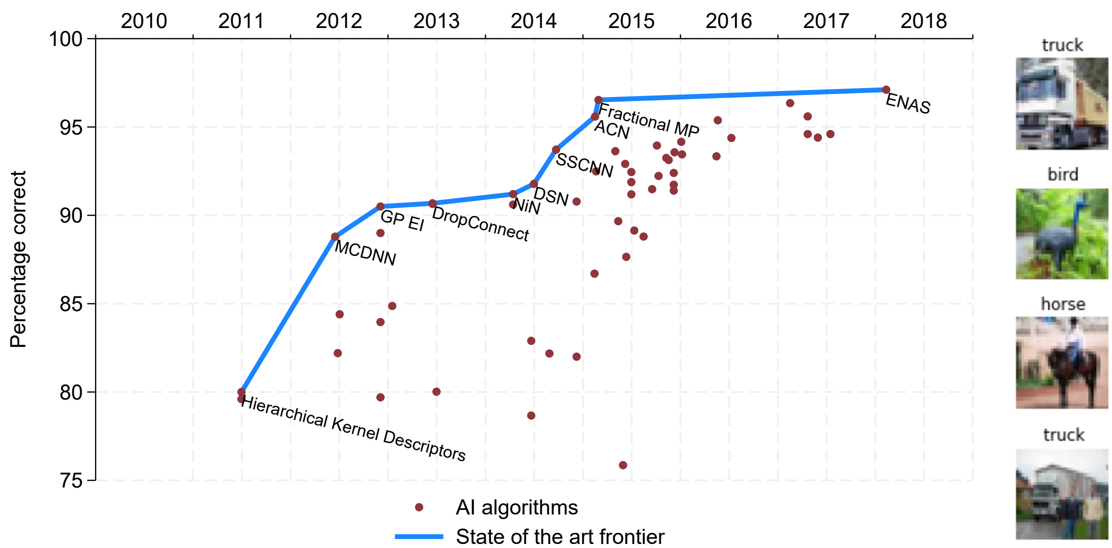


Figure 1: AI Performance and Benchmark Examples: CIFAR-10 Image Recognition

Notes: The blue line shows the state-of-the-art (SOTA) performance frontier before re-scaling. Named points indicate the best-performing AI models at the time of publication. The y-axis reports classification accuracy (i.e., percentage of correctly labelled images). On the right are sample CIFAR-10 images (32×32 pixels) along with their labels.

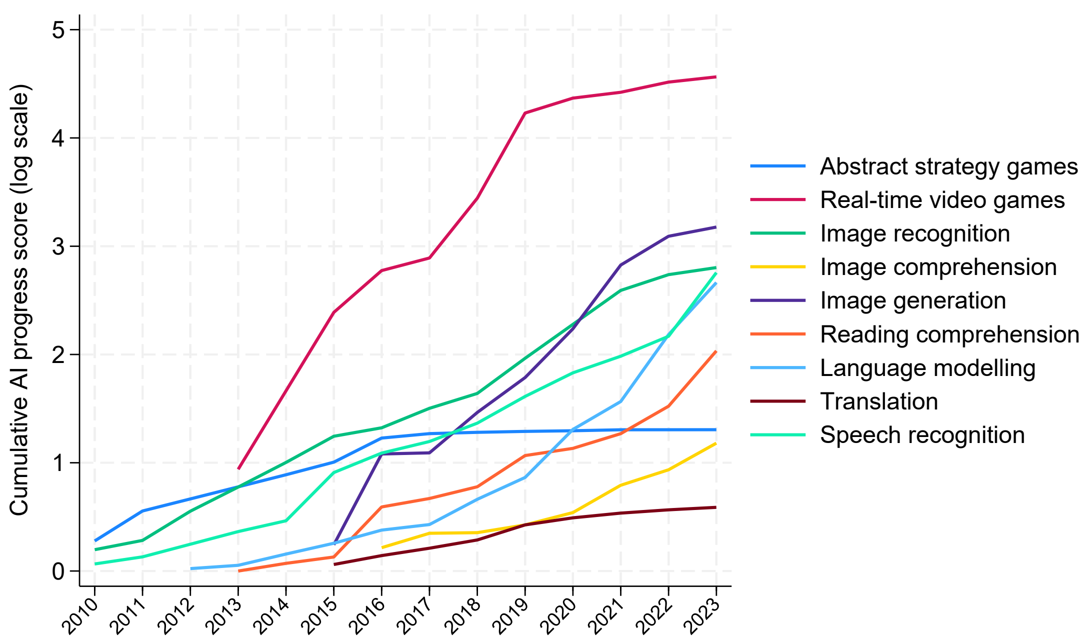


Figure 2: Estimated Progress Over Time by AI Application

Notes: Progress curves for each AI application are derived from the underlying benchmarks, using the average slope of benchmark frontiers by year. The resulting application-level progress measures are subsequently linked to worker abilities through the *mapping matrix*.

For each occupation o and ability j, we compute a relevance score r_{oj} following Felten *et al.* (2021), by multiplying O*NET's normalized scores of importance (i_{oj}) and level (l_{oj}), and rescaling so that the values sum to one within each occupation:[13]

$$r_{oj} = \frac{i_{oj} l_{oj}}{\sum_{j=1}^{52} i_{oj} l_{oj}} \tag{1}$$

This ensures comparability across occupations with different breadths of ability requirements.

To link AI applications i to worker abilities j, we use the 9×52 mapping matrix from Felten *et al.* (2018), which assigns a relatedness score $x_{ij} \in [0, 1]$ between each application and ability, based on expert assessment.[14] The full matrix is shown in the Online Appendix.

Cognitive abilities are most strongly linked to AI applications, followed by sensory abilities, while physical and psychomotor abilities show limited connections, except for video games, which notably combine perception and physical action. This pattern reflects AI research priorities from 2010 to 2023, which emphasised cognitive over robotic progress. As Hinton (2024) remarked, "these things aren't yet very good at physical manipulation. That will probably be the last thing they're very good at". Another famed AI researcher, Ilya Sutskever, attributed this to the difficulty of obtaining data for robotics (Patel, 2023).

Following Felten *et al.* (2018), we integrate AI progress into the mapping matrix by weighting each application's relatedness score x_{ij} with its annual progress Δp_{it}. Summing across applications yields the exposure change for ability j in year t: $\Delta e_{jt} = \sum_{i=1}^{9} \Delta p_{it} x_{ij}$. We then compute the exposure of occupation o in year t by weighting ability-level exposure changes Δe_{jt} by the relevance of each ability to the occupation, r_{oj}: $\Delta e_{ot} = \sum_{j=1}^{52} \Delta e_{jt} r_{oj}$.

While the 52 O*NET abilities provide a broad view of occupational content, they do not

[13] Importance and level are rated on 1–5 and 1–7 scales, respectively. See the Online Appendix for details.

[14] We use the Felten *et al.* (2018) matrix, derived from expert judgment, as it better reflects past AI capabilities. The Felten *et al.* (2021) matrix, crowd-sourced via MTurk, is designed to capture likely future uses.

fully capture the role of social interaction. To address this, we incorporate the information on social skills in O*NET, such as the occupational importance of persuasion and social perceptiveness, by constructing an index S_o of social intensity for each occupation.[15] This index is computed by interacting level and importance scores for each occupation o and social skill s, summing across social skills, and normalising by the highest observed value:

$$S_o = \frac{\sum_{s=1}^{6} i_{os} l_{os}}{\max_{o \in O} \left(\sum_{s=1}^{6} i_{os} l_{os} \right)} \tag{2}$$

We assume that social tasks are more difficult to automate, in line with Deming (2017) who finds rising demand and wages in jobs combining social and cognitive skills, and with Agrawal *et al* (2019), who argue AI excels at prediction but not judgment. To down-weight AI exposure in socially intensive jobs, we apply a non-linear transformation:

$$\Delta DAIOE_{ot} = \left(\Delta e_{ot} \times \frac{1 - S_o + \delta}{\max_{o \in O}(1 - S_o + \delta)} \right)^2 \tag{3}$$

Here, δ governs the strength of the adjustment. With $\delta = 2$, the exposure score of the most social occupation is reduced by 31 percent. This adjustment reflects the view—supported by the literature—that social interaction is a dimension of work less amenable to AI substitution.[16]

We then define the cumulative exposure measure as:

$$DAIOE_{ot} = \sum_{2010}^{t} \Delta DAIOE_{ot} \tag{4}$$

This captures the accumulation of exposure since 2010, when AI benchmarks began showing systematic progress.[17] Sub-measures for each AI domain, $DAIOE_{oti}$, are constructed

[15]For the full list of social skills, see the Online Appendix. Our social skills measure is highly correlated with that used by Deming (2017); see the Online Appendix for a comparison.

[16]The squaring serves to enhance the variation between occupations over time.

[17]The resulting measure is an ordinal variable, rank-ordering the occupations according to their exposure

analogously, enabling analysis across domains such as vision, language, and games.[18]

In sum, we construct a dynamic, subdomain- and year-specific index of occupational AI exposure that captures both automation and complementarity channels without attempting to disentangle them.

2.2. Descriptive Analysis

Having constructed the DAIOE measure, we now turn to a descriptive analysis. Table 2 presents the five most and least AI-exposed occupations in 2023 across three versions of the measure: overall AI exposure, and exposure to two specific subdomains—image generation and speech recognition.[19] While there is substantial overlap, with knowledge-intensive occupations typically ranking highest and physically demanding ones lowest, notable differences emerge across subdomains. For instance, telemarketers are ranked the most exposed occupation to speech recognition but much lower for image-based AI.[20] Similarly, detailed results show that, for example, dental nurses appear in the bottom half for most subdomains, but rise above the median in exposure to AI in video games, likely reflecting the inclusion of patient interaction simulations and training modules that use gaming technologies. These examples highlight the heterogeneity of AI exposure across subdomains, even within a single occupation.

Figure 3 traces the evolution of AI exposure from 2010 to 2023 for seven selected occupations, positioned across the exposure distribution. The figure reveals a widening dispersion over time and a clear acceleration of AI progress starting around 2012, coinciding with the rise of deep learning. A pivotal moment was the introduction of AlexNet in the 2012 ImageNet

to AI. For occupation-level analysis, we typically standardise the measure, subtracting the mean (0) and dividing by one standard deviation, while for firm-level estimations, we standardise the measure, using the firm-year-level mean and standard deviation in the respective country.

[18] While Felten *et al.* (2023) expands on Felten *et al.* (2021) by providing exposure measures specifically for language modelling and image generation, we develop measures for all nine AI applications, and one for generative AI, combining language modelling and image generation.

[19] Equivalent tables for all nine applications are available in the Online Appendix.

[20] In image generation and recognition, telemarketers are ranked at the 37th and 3rd percentiles, respectively.

TABLE 2

Most and Least Exposed Occupations by AI Application (2023)

	DAIOE overall	
Rank	*Most Exposed*	*Least Exposed*
1	Proofreaders and Copy Markers	Dancers
2	Mathematicians	Fitness Trainers and Aerobics Instructors
3	Mathematical Technicians	Athletes and Sports Competitors
4	Statisticians	Choreographers
5	Technical Writers	Forest Firefighters
	Image Generation	
Rank	*Most Exposed*	*Least Exposed*
1	Poets, Lyricists and Creative Writers	Slaughterers and Meat Packers
2	Mathematicians	Graders and Sorters, Agricultural Products
3	Graphic Designers	Fitness Trainers and Aerobics Instructors
4	Fine Artists, Including Painters, Sculptors, and Illustrators	Orderlies
5	Technical Writers	Flight Attendants
	Speech recognition	
Rank	*Most Exposed*	*Least Exposed*
1	Telemarketers	Dancers
2	Proofreaders and Copy Markers	Structural Iron and Steel Workers
3	Credit Checkers	Manufactured Building and Mobile Home Installers
4	Telephone Operators	Pressers, Textile, Garment, and Related Materials
5	Medical Transcriptionists	Reinforcing Iron and Rebar Workers

Notes: The table lists the occupations with the highest and lowest exposure scores in 2023 based on the DAIOE measure. Results are shown for overall AI exposure and two specific AI subdomains: image generation and speech recognition. Rankings are based on 8-digit O*NET-SOC codes. Equivalent rankings for all nine subdomain measures are available in the Online Appendix.

competition, which marked a leap forward in image recognition and helped catalyse rapid advances in multiple AI subdomains.

Figure 4 illustrates the distribution of AI exposure across major occupational groups. While there is considerable variation within each group, a clear pattern emerges: white-collar occupations (ISCO groups 1–4) exhibit significantly higher average exposure to AI. Specifically, groups 1–3—comprising Managers, Professionals, and Technicians and Associate Professionals—typically require higher education qualifications, as defined by ILO in the ISCO. Group 4, Clerical Support Workers, although generally not requiring tertiary education, is predominantly associated with office-based tasks.[21]

[21] For more information on the skill levels in ISCO, see: https://www.ilo.org/public/english/bureau/stat/isco/isco08/.

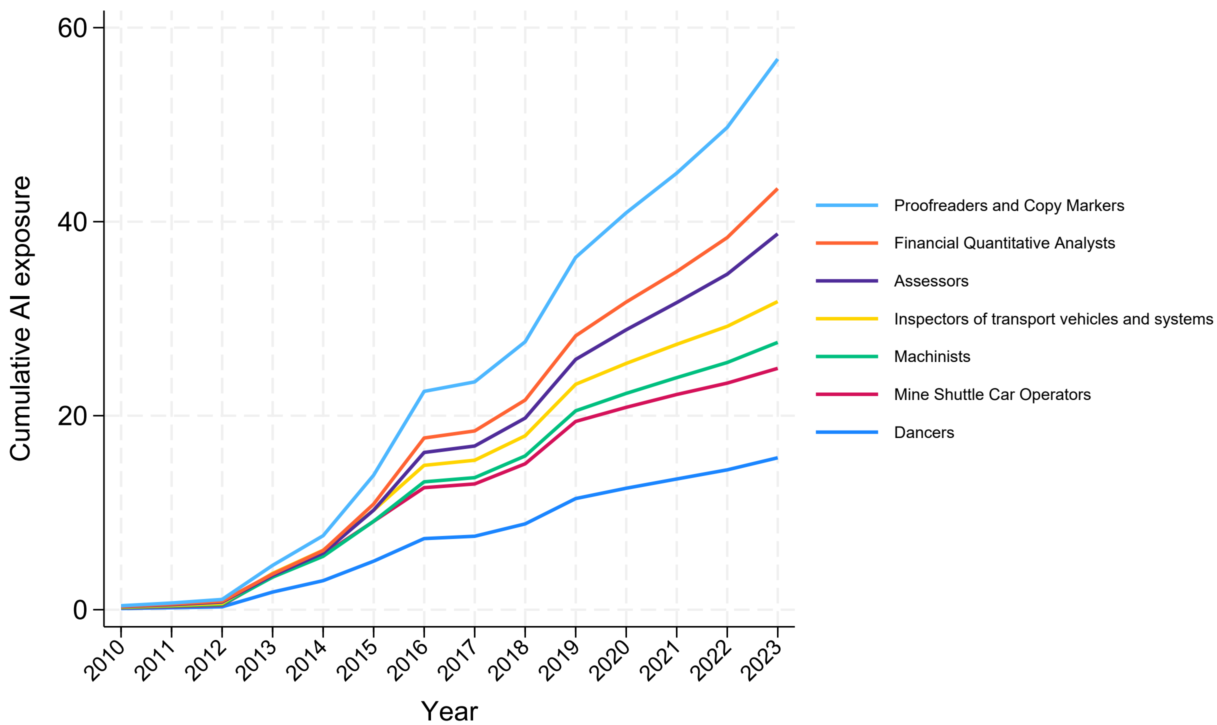


Figure 3: DAIOE Trajectories Over Time for Selected Occupations

Notes: The figure shows AI exposure (DAIOE) over time for seven occupations, selected to represent the 0th, 10th, 25th, 50th, 75th, 90th, and 100th percentiles of the 2023 DAIOE distribution.

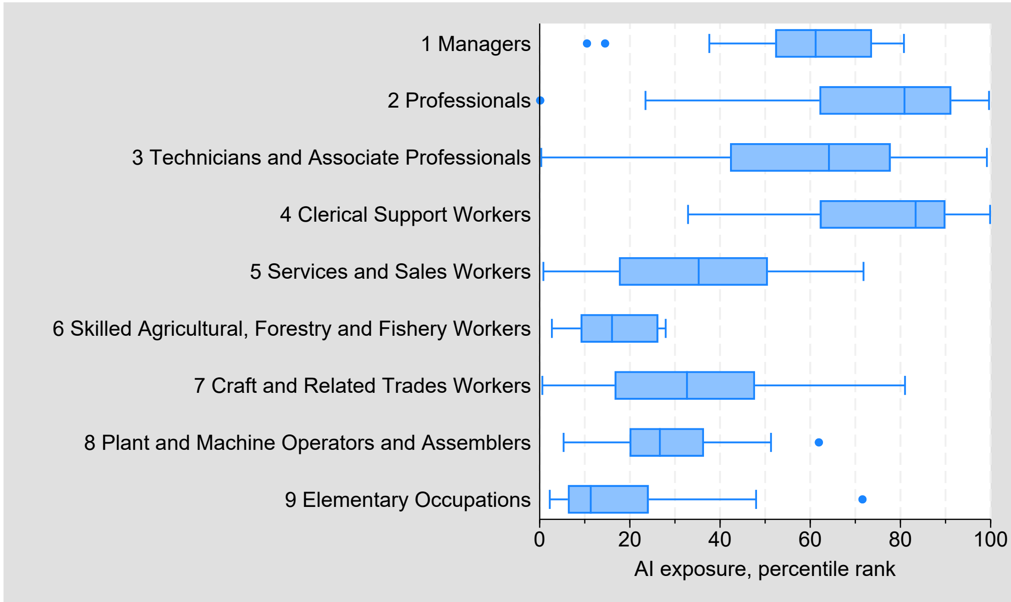


Figure 4: AI Exposure by Broad Occupation Group

Notes: The box plot displays the distribution of percentile rankings for occupations in the DAIOE measure in 2023, grouped by one-digit ISCO-08 occupation categories. Within each box, vertical lines mark the 25th, 50th (median), and 75th percentiles. Whiskers represent the full observed range, unless outliers are present, in which case they are shown as individual points. Exposure could not be calculated for group 0 (Armed Forces Occupations) due to the absence of relevant O*NET data.

2.3. Robustness Analysis

As an initial robustness check, we compare the DAIOE measure for the year 2023 with several prominent alternative measures from the literature: the backward-looking measure from Felten *et al.* (2018) (FRS18); the forward-looking AIOE from Felten *et al.* (2021) (FRS21); the patent-and-task-based measure by Webb (2019) (WEB19); the probability-of-computerization measure from Frey and Osborne (2017) (FO17); and the task-based measure of exposure to large language models from Eloundou *et al.* (2024) (Open24)[22]. Figure 5 displays the correlations between these measures, with one observation per SOC-2010 occupation. As anticipated, DAIOE is strongly positively correlated with both FRS18 and FRS21, reflecting the methodological similarities. There is also a strong positive correlation with Open24, even though they use a different, task based modelling approach. Open24

[22]We focus on their measure E1+E2, which is intended to reflect medium-term exposure, which we think is most consistent with the interpretation of DAIOE. We use their human-labelled exposure, as we interpret it as their primary version, rather than the AI-labelled one.

focus on exposure to large language models, and it is extra correlated with our sub-index for language modelling, with a correlation coefficient of 0.84. There is also a positive but weaker correlation between DAIOE and WEB19. Interestingly, FO17 shows negative correlations with all other measures, particularly with FRS18 and FRS21.

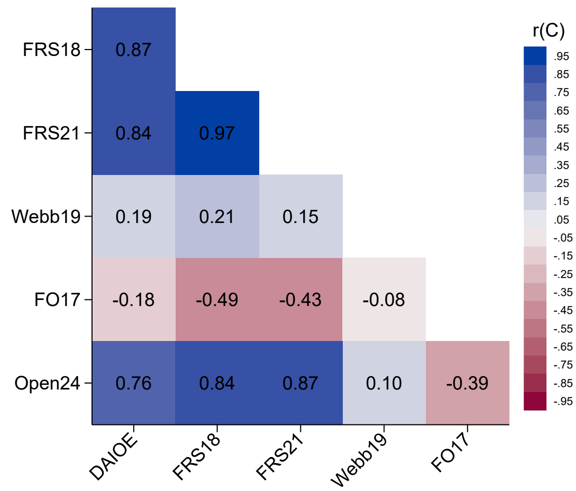


Figure 5: Correlation Between DAIOE and Alternative AI Exposure Measures

Notes: Correlations between DAIOE and alternative AI exposure measures. FRS18 = Felten *et al.* (2018); FRS21 = Felten *et al.* (2021); WEB19 = Webb (2019); FO17 = Frey and Osborne (2017); Open24 = Eloundou *et al.* (2024), specifically the human-labelled E1+E2 measure. One observation per SOC-2010 occupation. *, **, and *** indicate statistical significance at the 5%, 1%, and 0.1% levels, respectively.

To explore the relationship between AI exposure and occupational characteristics, Table 3 reports estimates from a simple regression model. The dependent variable is the standardised value (in standard deviations) of one of the AI exposure measures, while the explanatory variables are the occupation's O*NET scores for social skills, cognitive abilities, and physical abilities. These scores range from 0 to 1, with 1 assigned to the occupation with the highest value in each category. As in Figure 5, there is one observation per SOC-2010 occupation.

The results from Column 1 indicate that occupations with high cognitive demands and limited social or physical requirements tend to be the most exposed to AI, according to the

TABLE 3
AI Exposure Measures and Occupational Characteristics

	(1) DAIOE	(2) FRS18	(3) FRS21	(4) Webb19	(5) FO17	(6) Open24
Social skills	-3.097***	0.232**	1.040***	-2.593***	-2.819***	0.691***
	[-25.10]	[2.80]	[12.05]	[-8.77]	[-12.72]	[3.66]
Cognitive abilities	3.315***	3.077***	1.451***	4.468***	-3.163***	1.593***
	[19.59]	[27.11]	[12.25]	[11.08]	[-10.57]	[6.17]
Physical abilities	-5.894***	-4.642***	-5.074***	0.238	-0.744***	-4.138***
	[-53.16]	[-62.42]	[-65.38]	[0.90]	[-3.78]	[-24.40]
R^2	0.849	0.932	0.926	0.145	0.561	0.648
Observations	772	772	772	771	685	768

Notes: OLS regressions are estimated for DAIOE and several alternative AI exposure measures against broad occupational characteristics. FRS18 = Felten *et al.* (2018); FRS21 = Felten *et al.* (2021); WEB19 = Webb (2019); FO17 = Frey and Osborne (2017). To ensure comparability, all exposure measures are standardised (mean = 0, SD = 1). The explanatory variables are based on O*NET level × importance scores, summed across abilities and normalised by the value for the highest-scoring occupation in each category. For example, an occupation scoring half as high on physical abilities as the top-ranked occupation would receive a score of 0.5. Coefficients reflect the change in standard deviations of the exposure measure when moving from zero to the highest observed ability score in each category. One observation per SOC-2010 occupation. t-statistics are reported in brackets. *, **, and *** indicate significance at the 5%, 1%, and 0.1% levels, respectively.

DAIOE measure in Column 1. The high R^2 suggests that these three occupational dimensions explain a substantial share of the cross-occupational variation in DAIOE scores. This pattern aligns with the top-exposed occupations listed in Table 2; for instance, proofreaders and copy markers and mathematicians are occupations that fit this profile. In contrast, the least exposed occupations, such as dancers and firefighters, tend to involve highly physical and/or social tasks.

A comparison of the DAIOE measure with alternative AI exposure metrics in Columns 2–5 reveals broadly consistent patterns in how occupational characteristics relate to technology exposure. Across all measures—except for FO17—cognitive abilities are positively associated with exposure, indicating that occupations requiring higher cognitive skills are generally more susceptible to AI technologies. Notably, while DAIOE, WEB19, and FO17 suggest that occupations with higher social skill requirements are less exposed to technology, the measures FRS18 and FRS21 indicate a positive relationship between social skills and AI exposure. Regarding physical abilities, all measures—except WEB19, where the relationship is statistically insignificant—demonstrate a negative association with AI exposure, suggesting that physically intensive occupations are less likely to be affected by AI advancements.

As an external check on whether DAIOE indeed reflects real-world AI adoption, we correlate occupational DAIOE scores with the share of AI-related job ads in Sweden (see Online Appendix I, Figures I1–I2). We find a clear, positive relationship: occupations with higher DAIOE percentiles also exhibit a larger fraction of online listings demanding AI-related skills. This supports interpreting DAIOE as capturing meaningful variation in AI technologies to which firms respond.

3. Conceptual Framework

To guide our empirical analysis of AI's labour-market impacts via DAIOE, we extend the Acemoglu–Restrepo (2018) continuum-of-tasks structure in three dimensions:[23] a multidimensional, abilities-based index of occupational complexity; a codifiability (standardisation) channel that raises labour productivity; and social-skill intensity that alters substitution between labour and capital.

Firms produce output Y by aggregating a continuum of occupation-level modules $o \in [0, 1]$, ordered by complexity, via a CES aggregator:

$$Y = \left(\int_0^1 V(o)^{\frac{\sigma-1}{\sigma}} \, \mathrm{d}o \right)^{\frac{\sigma}{\sigma-1}}, \tag{5}$$

where $\sigma > 0$ and $V(o)$ is the net output of occupation o.

Each occupation o requires a bundle of complementary abilities. We measure its complexity as

$$R_o = \sum_{j=1}^{52} i_{oj} \, l_{oj}, \tag{6}$$

where i_{oj} and l_{oj} are ONET importance and level scores. We assume $\gamma(R)$, the labour-

[23]We also draw on Aghion *et al* (2019b) and Bessen *et al* (2022).

productivity function, satisfies $\gamma'(R) > 0$ and $\gamma''(R) < 0$.

Net output $V(o)$ can be produced by labour or by capital, where capital here represents an AI application. Formally,

$$V(o) \;=\; \alpha\, K(o) \;+\; \gamma\big(R_o, \tilde{W}, \tilde{S}\big)\, L(o), \tag{7}$$

where α denotes capital (AI) productivity, which evolves with technological frontier gains. Labour productivity is given by $\gamma(R_o, \tilde{W}, \tilde{S})$, a function that depends on the routinisation index R_o of occupation o, the degree of codifiability or standardisation of the required abilities ($\tilde{W}$), and the social-skill intensity of the occupation ($\tilde{S}$).

Cost-minimisation yields

$$C(o) = \min\Big\{ \tfrac{w_k}{\alpha},\; \tfrac{w_l}{\gamma(R_o, \tilde{W}, \tilde{S})} \Big\}\; V(o), \tag{8}$$

with factor prices w_k and w_l. Occupation o is automated if $\frac{w_k}{\alpha} < \frac{w_l}{\gamma(R_o, \tilde{W}, \tilde{S})}$, defining a threshold R^* via

$$\frac{w_k}{\alpha} \;=\; \frac{w_l}{\gamma(R^*, \tilde{W}, \tilde{S})}. \tag{9}$$

We model codifiability by

$$\tilde{W} = \prod_{j=1}^{52} z_{j,0}, \tag{10}$$

where $z_{j,0} \in [0,1]$ is the initial AI exposure of ability j. Social-skill intensity $\tilde{S} \in [0,1]$ scales resistance to automation.[24]

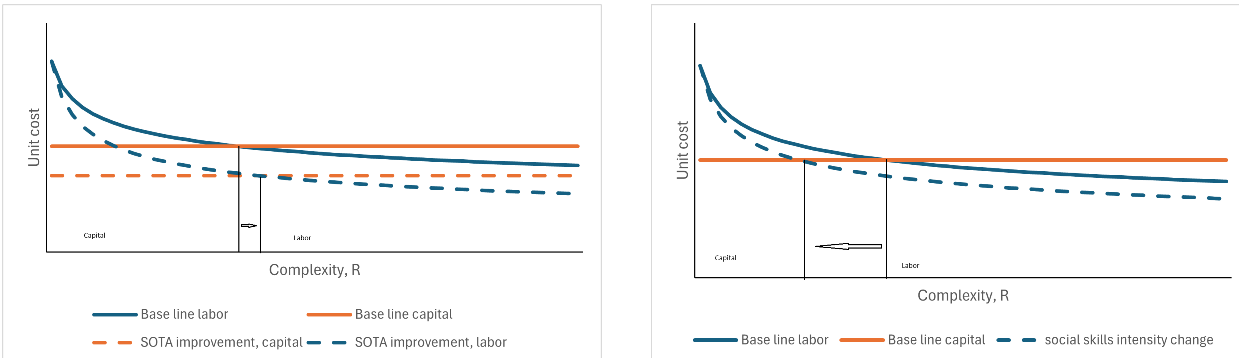
A comparative static shows

$$\frac{\partial R^*}{\partial \tilde{W}} < 0, \quad \frac{\partial R^*}{\partial \tilde{S}} > 0, \tag{11}$$

so greater codifiability lowers—and richer social-skill intensity raises—the automation thresh-

[24]One can microfound $\tilde{S}$ as moderating the elasticity of substitution between human and machine inputs.

old. As illustrated in Figure 6, panel (a) frontier gains shift down the capital-cost curve (and, via higher $\tilde{W}$, rotate the labour-cost curve), while panel (b) higher $\tilde{S}$ steepens the labour-cost curve, reducing automability.



(a) Frontier gains lower capital cost and raise $\tilde{W}$. (b) Higher $\tilde{S}$ steepens the labour-cost curve.

Figure 6: Comparative statics of automation and complementarity

Notes: Panel (a) shows how AI-performance gains shift down the capital-cost curve and can rotate the labour-cost curve via increased $\tilde{W}$. Panel (b) shows how higher social-skill intensity ($\tilde{S}$) steepens the labour-cost curve, reducing automability.

Equations (5)–(11) capture two opposing AI channels: substitution via cheaper capital ($\alpha \uparrow$) and complementarity via ability standardisation ($\tilde{W} \uparrow$), both moderated by social-skill intensity ($\tilde{S}$). In our empirical work, DAIOE proxies frontier gains and codifiability, while moderated by social-skill scores, allowing us to test heterogeneous labour-demand responses across occupations and AI subdomains.

4. Empirical approach

To investigate whether and how different types of AI affect labour demand over time, we apply our Dynamic AI Occupational Exposure (DAIOE) measure to comprehensive micro-data from three countries. This section introduces the selected countries, describes the micro-data, and outlines the regression model used to estimate labour demand.

4.1. A Primer on the Three Economies

We focus on Denmark, Portugal, and Sweden—three relatively open economies that differ markedly in labour market characteristics, industrial structures, and digital adoption levels (see Figures J1-J2 of the Online Appendix). These differences provide a valuable context for examining both general patterns and country-specific impacts of AI on labour markets.[25]

Denmark and Sweden are characterised by flexible labour markets combined with strong employment security, a model often referred to as "flexicurity" . These countries have maintained an integrated labour market for nearly 70 years and share many features of the Nordic welfare model. In contrast, Portugal exhibits a more rigid labour market structure, with lower employment rates, labour market participation, and post-secondary education attainment.

In terms of digital intensity, Denmark and Sweden lead with the highest proportions of digitally intensive enterprises in the EU. Portugal, however, lags in digital intensity and productivity. Compared to Denmark and Sweden, Portugal is more focused on manufacturing industries.

Regarding AI adoption, recent Eurostat data indicate that in 2024, approximately 25.2 percent of enterprises in Sweden and 27.6 percent in Denmark employed AI technologies, compared to just 9 percent in Portugal (Eurostat, 2024). This disparity underscores the varying degrees of AI integration across these economies.

In terms of economic structure, Portugal maintains a stronger focus on manufacturing industries, whereas Denmark and Sweden are more services-oriented. This structural difference may influence the adoption and impact of AI technologies across these economies.

[25]The countries also differ in economic and geographic size as well as location. Sweden, Denmark and Portugal jointly account for 7% and 5% of the EU GDP and population. Geographically, Sweden is the third largest EU country, while Portugal and Denmark are ranked 12 and 21 in size. Together, the three countries cover three of the four European climate zones, from Nordic to Mediterranean climate, representing vastly different conditions, e.g., for the primary sector.

These distinctions among the three countries provide a context for examining how AI technologies may influence labour demand, both in terms of shared trends and country-specific dynamics. The deep digital penetration and services-orientation of Sweden suggests a high baseline of AI complementarities—leaving somewhat less room for incremental up-skilling—whereas Portugal's lower starting point offers greater scope for compositional shifts as frontier AI performance improves.

4.2. Register Data

We use longitudinal register data from Denmark, Portugal, and Sweden to analyse the relationship between AI exposure and labour demand. The data span from 2010 to the most recent available year: 2021 for Denmark and Portugal, and 2020 for Sweden. Each dataset includes unique identifiers for firms and workers, allowing us to exactly match employees to their employers in each year and construct linked employer–employee data (LEED) for each country. This precise, administrative matching contrasts with studies that rely on statistical matching across surveys or imputation techniques thereby minimising measurement error in key variables such as workforce composition and firm characteristics. Due to legal restrictions, the data cannot be pooled across countries; we therefore conduct identical analyses separately.

The LEED data include harmonised information across countries. For workers, we observe education, occupation, and employment status. For firms, we use information on age, location, industry, employment, capital inputs, outputs, foreign trade, and ownership. All monetary variables are deflated to 2015 levels and converted to USD.[26]

We link the DAIOE measure to firm-level employment using administrative records on firms' occupational composition at the 4-digit level in the initial year. To address potential endogeneity—namely, that a firm's occupational structure may itself influence the likelihood of adopting AI—we focus on baseline workforce composition. Firms with a higher initial

[26]Details on country-specific sources are provided in the Online Appendix.

share of employees in occupations with high expected exposure are thus assigned greater AI exposure.[27] We compute the measure both using fixed occupational shares from the initial year.

Table 4 presents descriptive statistics for the estimation samples. Firms in all three countries are predominantly small, bordering on micro-enterprises. Danish and Swedish firms are similar in size and structure, while Portuguese firms tend to be smaller, less capital- and human-intensive, and less engaged in trade and foreign ownership, consistent with the macro-level differences described earlier.

The workforce composition by occupational group and industry is presented in Tables A1 and A2.[28] Across the three countries, the distribution of workers among blue-collar, low-skilled white-collar, and high-skilled white-collar categories is approximately even, with each group comprising about one-third of the workforce. Denmark and Sweden exhibit similar distributions of blue-collar and low-skilled white-collar workers. However, Sweden distinguishes itself with a higher proportion of high-skilled white-collar workers. In contrast, Portugal has the largest shares of blue-collar and low-skilled white-collar workers and the smallest share of high-skilled white-collar workers.

In terms of industrial distribution, the countries show notable differences. Sweden has a smaller share of employment in manufacturing and a relatively larger presence in business services and construction. Conversely, Portugal has the highest share of workers in manufacturing but lower proportions in information and communication as well as in business services.

To provide a stylized view of how firms with initially low versus high exposure to AI have evolved in terms of their occupational structure, we have analysed the shares of 1-digit ISCO occupational groups in the first and last years of the panel, stratified by the first

[27] For occupations with missing exposure scores, we compute the weighted average based on the remaining occupations.

[28] Blue-collar workers are defined as those in ISCO-08 categories 6–9, low-skilled white-collar workers as categories 4–5, and high-skilled white-collar workers as categories 1–3.

Table 4

Firm-Level Descriptive Statistics

Variable	Mean	Median	S.D.
		Denmark	
Workforce size	36.07	11.98	203.61
DAIOE measure	12.47	13.02	9.20
Robot measure	0.61	0.55	0.39
Software measure	0.44	0.42	0.16
Offshorability measure	0.50	0.52	0.11
Sales (1000 USD)	15,319.31	2,702.05	154,938.40
Value Added (1000 USD)	4,227.91	1,012.65	46,946.81
Skilled worker share	0.23	0.13	0.26
Capital intensity (1000 USD)	276.87	20.81	15,815.34
Firm age	18.08	14.00	15.21
Foreign-owned (D)	0.10	0.00	0.30
Export value (1000 USD)	5,055.20	0.00	104,219.70
Import value (1000 USD)	4,053.46	2.88	91,027.12
		Portugal	
Workforce size	21.90	7.57	137.46
DAIOE measure	12.40	13.12	9.01
Robot measure	0.62	0.55	0.41
Software measure	0.43	0.41	0.17
Offshorability measure	0.51	0.52	0.11
Sales (1000 USD)	4,617.81	679.90	6,2316.89
Value Added (1000 USD)	1,113.15	234.32	11,325.59
Skilled worker share	0.16	0.00	0.24
Capital intensity (1000 USD)	33.08	9.14	204.38
Firm age	19.19	16.00	14.55
Foreign-owned (D)	0.04	0.00	0.19
Export value (1000 USD)	1,003.66	0.00	22,619.44
Import value (1000 USD)	903.45	0.00	20,516.52
		Sweden	
Workforce size	34.41	11.00	218.45
DAIOE measure	11.38	12.83	8.38
Robot measure	0.59	0.55	0.37
Software measure	0.46	0.43	0.14
Offshorability measure	0.50	0.50	0.10
Sales (1000 USD)	13,259.40	1,987.37	157,357.36
Value Added (1000 USD)	3,622.40	827.43	33,164.96
Skilled worker share	0.20	0.10	0.24
Capital intensity (1000 USD)	200.52	20.77	5,572.92
Firm age	17.02	16.00	9.53
Foreign-owned (D)	0.08	0.00	0.27
Export value (1000 USD)	3,185.90	0.00	107,780.64
Import value (1000 USD)	2,797.16	0.00	71,046.53

Notes: Summary statistics for the full firm-year estimation sample in each country. The DAIOE measure (non-standardised) is the weighted average AI exposure of the firm, based on firm-specific 4-digit ISCO08 occupational shares from the first year of observation. Robot and Software measures are constructed analogously, using occupational exposure scores from Webb (2019); the Offshorability measure uses Autor and Dorn (2013). Workforce size is measured in full-time equivalents for Denmark and Portugal; skilled workers have a tertiary education. Capital intensity is the book value of physical capital per worker. Foreign-owned indicates firms controlled from abroad. All monetary values are in 1,000 USD. Sample sizes: Denmark: 51,233; Portugal: 126,265; Sweden: 95,512.

and fourth quartiles of AI exposure (see table A3). Firms with high initial exposure to AI consistently employ a substantially larger share of workers in occupational groups 1–4 (white-collar categories) compared to firms in the lowest exposure quartile.[29]

We observe heterogeneous trends between countries. In Denmark, the most exposed firms exhibit a general shift toward upskilling. In contrast, in Portugal, the most exposed firms have seen increases in some blue-collar occupations, suggesting a more mixed pattern.

A consistent finding across all three countries is that the least exposed firms are increasing their employment in ISCO-08 category 3 (technicians and associate professionals), while the most exposed firms are reducing their shares in this group. This trend is most pronounced in Sweden, where the share of workers in category 3 among the most exposed firms declined by 5.9 percentage points, reaching 24.9 percent in the final year. This reduction may indicate that AI systems are increasingly replicating the cognitive abilities characteristic of mid-level white-collar occupations, such as pattern recognition, rule-based decision-making, and structured communication. As AI technologies advance in these domains, they enable the substitution of human labour in roles traditionally performed by technicians and associate professionals, leading to a decline in demand for these occupations. These patterns align with recent findings in the literature (e.g., Babina *et al*, 2023).

4.3. Empirical Specification

To investigate the relationship between AI exposure and firm-level labour demand, we estimate the following baseline specification:

$$L_{ct} = \beta_0 + \beta_{AI} AI_{ct-1} + \mathbf{X'}_{ct-1}\boldsymbol{\beta_X} + \mathbf{D'}_{ht}\boldsymbol{\beta_{ht}} + \mathbf{D'}_{mt}\boldsymbol{\beta_{mt}} + \epsilon_{ct} \tag{12}$$

Here, $\log L_{ct}$ denotes the natural logarithm of employment for company c at time t. The

[29] The least exposed firms are concentrated in construction and trade, while the most exposed are primarily in other business services, including consulting and advertising.

primary variable of interest, AI_{ct-1}, represents the company's exposure to AI in the previous period, calculated as the weighted average of occupational AI exposure scores (DAIOE) based on the firm's baseline occupational composition at the ISCO-08 four-digit level.[30] To facilitate interpretation, AI_{ct-1} is standardised at the firm-year level within each country, using the mean and standard deviation of the respective firm-year distribution.

The vector $\mathbf{X'}_{ct-1}$ includes lagged firm-level control variables: log sales (as a proxy for firm size), physical capital intensity (capital-to-labour ratio), human capital intensity (share of employees with post-secondary education), firm age, export and import values, and a dummy for foreign ownership. Continuous variables are log-transformed. To ensure meaningful variation in occupational structures, we restrict the sample to firms with at least five employees. Regressions are weighted by firm employment in the first year of the panel.[31] Standard errors are clustered at the firm level.

The fixed effects $\mathbf{D'}ht$ and $\mathbf{D'}mt$ control for 3-digit industry-year and location-year heterogeneity, respectively.[32] This absorbs unobserved, time-varying shocks at the sectoral or local labour-market levels, such as changes in trade exposure, technological trends, or local demand shifts.

Our identification rests on variation in DAIOE that is driven entirely by shifts in aggregate AI benchmarks—developments over which individual firms have no control. By fixing each firm's occupational shares at baseline, we ensure that any change in its AI exposure score comes solely from external advances in AI capabilities, not from firms reshaping their workforce in anticipation of new technologies. In other words, we treat DAIOE's year-to-year movements as plausibly exogenous, isolating the effect of AI progress from firm-specific hiring or skill-mix decisions.

[30]For Denmark and Portugal, we use full-time-equivalent employment (and weights); substituting total employment yields only marginal differences.

[31]Our main results are qualitatively robust to excluding employment weights (Table A9-Table A10).

[32]For Denmark and Sweden, we use the municipality as the location, while for Portugal, the NUTS-2 classification of the EU.

To further isolate AI-specific effects, we add three analogous exposure indices: a robot-exposure index and a software-exposure index (both from Webb, 2019), plus an offshorability index (Autor and Dorn , 2013). Together with our core covariates, these alternative measures help ensure that estimated DAIOE effects are not confounded by other automation or globalisation drivers.

As a complement to our baseline panel specification—and to verify that results do not depend on the exact econometric setup—we also estimate a "long-difference" model comparing firm outcomes averaged over 2010–11 and 2020–21:

$$\Delta L_{c,(t_0 \to t_1)} = \beta_0 + \beta_{AI}\,\Delta AI_{c,(t_0 \to t_1)} + \mathbf{X'}_{c,t_0}\,\boldsymbol{\beta_X} + \mathbf{D'}_{h,t_0}\,\boldsymbol{\beta_{ht}} + \mathbf{D'}_{m,t_0}\,\boldsymbol{\beta_{mt}} + \Delta\varepsilon_{c,(t_0 \to t_1)}, \quad (13)$$

where $\Delta L_{c,(t_0 \to t_1)} = L_{c,t_1} - L_{c,t_0}$ denotes the change in log employment between the two periods, $\mathbf{X}_{c,t_0}$ includes the full set of baseline controls and industry- and location-year fixed effects measured at t_0, and $\Delta AI_{c,(t_0 \to t_1)} = AI_{c,t_1} - AI_{c,t_0}$ is computed using each firm's initial occupational shares.[33]

Our panel data, although rich, are unbalanced and span just over a decade, which limits the scope for additional high-dimensional firm-level fixed effects. By combining lagged covariates, two sets of time-varying fixed effects, and the long-difference check, we address potential confounding from both time-varying and time-invariant unobservables. Nevertheless, we interpret our findings as associations rather than definitive causal effects.

Finally, to explore heterogeneity across technologies, we re-estimate specification (12) separately for each of the nine AI subdomains in Figure 2, allowing us to trace how distinct AI capabilities map into labour-demand responses.

[33]Long-difference results qualitatively match our main panel estimates, also showing robust up-skilling across countries, and low-skill white-collar declines in Denmark and Portugal, while the high-skill white-collar coefficient loses statistical significance, highlighting some specification sensitivity in that margin (Table A8).

5. Regression Results

5.1. Employment Outcomes

We present our baseline regression results from estimating Equation 12 in Figure 7, with detailed estimates provided in Tables A4 and A5 in the Appendix. The figure visualises the coefficients and 95 percent confidence intervals for firm-level AI exposure on four firm-level employment outcomes across Denmark, Portugal, and Sweden. In addition to measuring the association between AI exposure and total employment, we examine heterogeneity across occupational groups by dividing a firm's workforce into three broad categories: "white-collar-high," encompassing occupations in ISCO-08 major groups 1 to 3; "white-collar-low," including groups 4 and 5; and "blue-collar," comprising groups 6 through 9.

Our baseline estimates show no significant link between AI exposure and total firm employment in any country (Denmark: 0.383; Portugal: 0.125; Sweden: 0.013). This null finding is consistent with our framework (Figure 6), where frontier AI gains both lower capital costs (the "substitution" channel) and raise labour productivity via standardisation (the "complementarity" channel), offsetting each other around the automation threshold R^* (Equation 9).[34]

However, the heterogeneity of AI exposure's association becomes apparent when analysing employment effects across occupational groups. As shown in Figure 7 and detailed in Table A5, the associations differ significantly across white-collar-high, white-collar-low, and blue-collar workers.

For white-collar-high workers (ISCO-08 major groups 1–3), we find consistently positive and statistically significant associations across all three countries. To gauge the magnitude, we consider the variation between firms at the 10^{th} and 90^{th} percentiles of AI exposure in the middle of our sample period (2015).[35] Compared to a firm at the 10^{th} percentile of AI

[34] Historical evidence suggests that general-purpose technologies often yield delayed net employment effects, unfolding over decades rather than years (Deming *et al.*, 2025).

[35] For Denmark, this variation corresponds to 21 percent of a standard deviation.

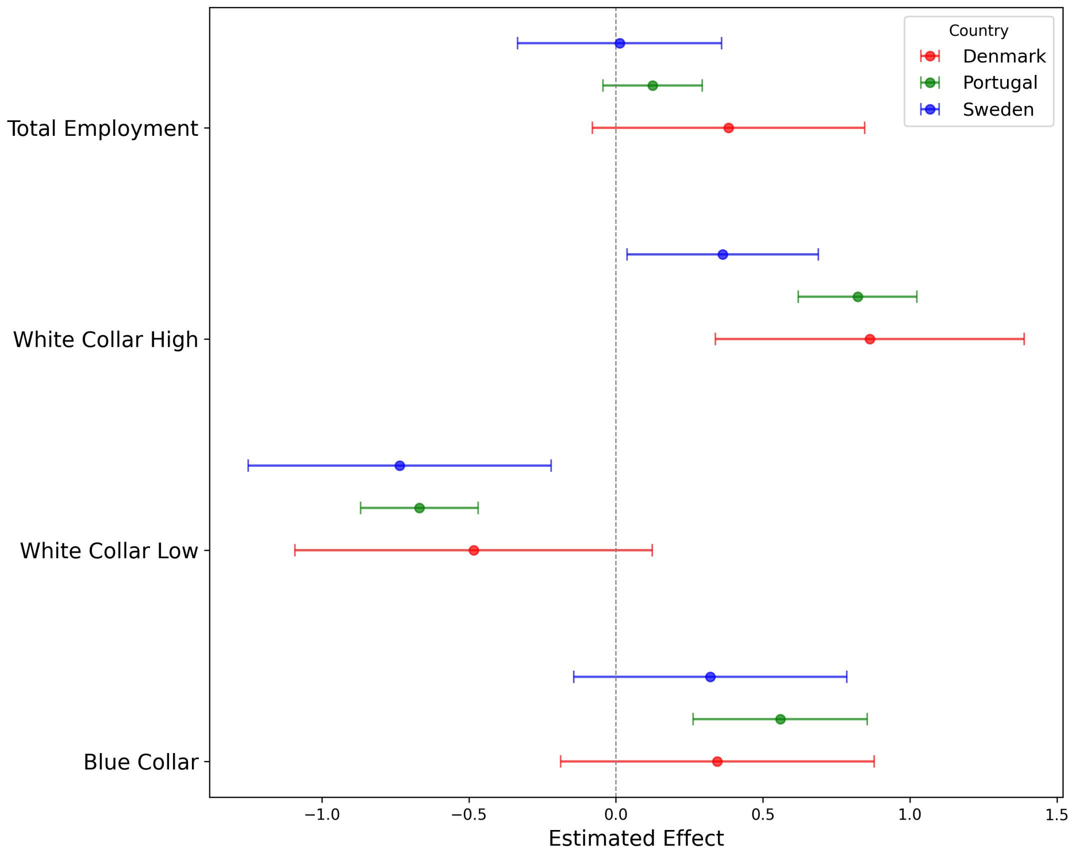


Figure 7: Employment Outcomes

Notes: The whisker plot depicts the estimated associations between AI exposure and different employment outcomes (total employment, white-collar high-skill, white-collar low-skill, and blue-collar occupations) across Denmark, Portugal, and Sweden. The horizontal bars represent the coefficient estimates for each country, with whiskers showing the 95% confidence intervals. The DAIOE measure is the standardised and weighted average AI exposure of the firm where the occupational composition is fixed at firm-specific baseline-year shares of 4-digit ISCO08 occupations. All regressions include fixed effects at 3-digit NACE industry-year and location-year levels. All regressors are lagged at $t-1$ except for the contemporaneous firm age. All continuous variables are in log form.

exposure, one at the 90$^{\text{th}}$ percentile exhibits 19 percent greater employment in white-collar-high occupations.

These results for high-skill white-collar occupations are best understood through our codifiability and social-skill channels. As AI benchmarks advance, greater standardization of core cognitive abilities (the $\tilde{W}$ channel) disproportionately raises labour productivity in the most complex jobs, rotating down the labour-cost curve and shifting the automation threshold R^* outward, while, at the same time, high-skill roles often rely on teamwork, judgment and leadership—dimensions captured by our social-skill intensity parameter $\tilde{S}$—which steepens the labour-cost curve and reinforces complementarities (Figure 6). Together, these mechanisms may drive the positive association we observe, rather than simply a displacement or scale effect.

For *white-collar-low* workers (ISCO-08 major groups 4 and 5), we observe a consistent negative association between firm-level AI exposure and employment, which is statistically significant for Portugal and Sweden. These occupations, which often involve routine cognitive tasks such as clerical and data processing work, appear more susceptible to automation through AI technologies. This pattern aligns with broader evidence indicating that AI-driven automation can displace jobs characterised by repetitive and predictable tasks. For instance, studies have highlighted that AI technologies are increasingly capable of performing tasks traditionally associated with clerical support, leading to potential reductions in employment within these roles (e.g., Gmyrek *et al.*, 2023).

For *blue-collar workers* (ISCO-08 major groups 6–9), the association between AI exposure and employment is generally muted across the three countries, with a statistically significant positive relationship observed only in Portugal. This pattern suggests that AI technologies, in contrast to previous technologies, such as industrial robots, are not uniformly targeted to tasks typical of blue-collar occupations, which often involve manual and physical activities. The observation aligns with the physical and context-specific nature of many blue-collar tasks, presenting challenges for AI.

Taken together, these occupational-group patterns align neatly with our framework's non-monotonic prediction (Section **??**): AI progress can lower the cost of capital (automation) for simpler, low-social-skill jobs yet raise labour productivity (complementarity) in more complex, codifiable roles, with social-skill intensity moderating where the substitution-complementarity threshold falls.

5.2. *Results by AI Subdomains*

Our framework implies that different advances along the AI frontier may lean more toward lowering the cost of capital (automation) or raising labour productivity (complementarity), so we should observe heterogeneous employment associations across subdomains. To test this, we re-estimate our baseline specification separately for each of the nine DAIOE indices, tracing how exposure to gaming, vision, language, and the other domains relates to total employment and to the shares of high-skill white-collar, low-skill white-collar, and blue-collar workers.

Figure 8 and Table A6 report these subdomain-specific estimates. The heatmap in panel (a) illustrates the coefficient on total employment, while panels (b)–(d) break the effects out by occupational group. As anticipated, some subdomains (e.g., video games) display positive links with overall employment and blue-collar shares—consistent with enhanced codifiability—whereas others (notably vision and language) tend to depress total employment and lower-skill shares, reflecting more direct automation pressures.

More specifically, each row in the heatmap (Figure 8) corresponds to one of the nine AI subdomains, while each column represents an employment outcome: total employment, white-collar-high, white-collar-low, and blue-collar occupations. Cell values denote coefficient estimates, while colour intensity reflects effect magnitude, with red and blue signifying positive and negative associations, respectively.[36]

Regarding total firm employment, the heatmap reveals that the association between AI exposure and employment varies across subdomains and countries. For several subdomains, coefficients are small and statistically insignificant, reflecting the limited aggregate associations depicted in Figure 7. However, certain subdomains, such as video games, exhibit positive and statistically significant associations with total employment, suggesting that AI technologies in this area may create new opportunities across multiple occupational groups

[36]The DAIOE measure is a standardised, weighted average of firm-level AI exposure (Equation 4), calculated using pre-sample firm-specific 4-digit ISCO08 occupational shares.

by enabling remote control and automation that require human oversight. This observation is consistent with our conceptual framework, which posits that expanding the range of abilities and enhancing the relevance of complementary skills can increase labour demand. Conversely, subdomains like image generation and language modelling—core areas of generative AI—display negative associations with total firm employment, indicating potential displacement effects. These findings highlight the importance of disentangling specific impacts across occupational groups to better understand the underlying mechanisms.

For white-collar-high workers, positive associations with AI exposure are observed across all three countries, though the magnitude varies by subdomain. Technologies such as image recognition and image compression exhibit the largest positive associations. Advancements in the AI subdomain of strategy games also show consistently positive associations with employment in professional and managerial occupations across Denmark, Portugal, and Sweden. These findings suggest a potential complementarity between AI and advanced human cognitive abilities required for tasks such as decision-making and strategic planning.

While the overall pattern is broadly consistent across countries, the comparatively smaller effect sizes observed in Sweden, relative to Denmark and Portugal, may relate to structural differences in labour markets and industry composition. As already mentioned, Sweden's economy is characterised by high productivity, strong digital intensity, and a concentration of employment in innovation-driven sectors and high-skilled occupations. These features suggest that Swedish firms may already operate at a high technological baseline, with a more limited scope for additional AI-driven complementarities. In contrast, firms in Denmark and Portugal—while similarly advanced—display greater heterogeneity in workforce composition and operate in sectors where AI may complement tasks more directly. This may help explain the somewhat stronger associations observed there between AI exposure and labour demand.

For white-collar-low workers, the associations with AI exposure vary considerably across

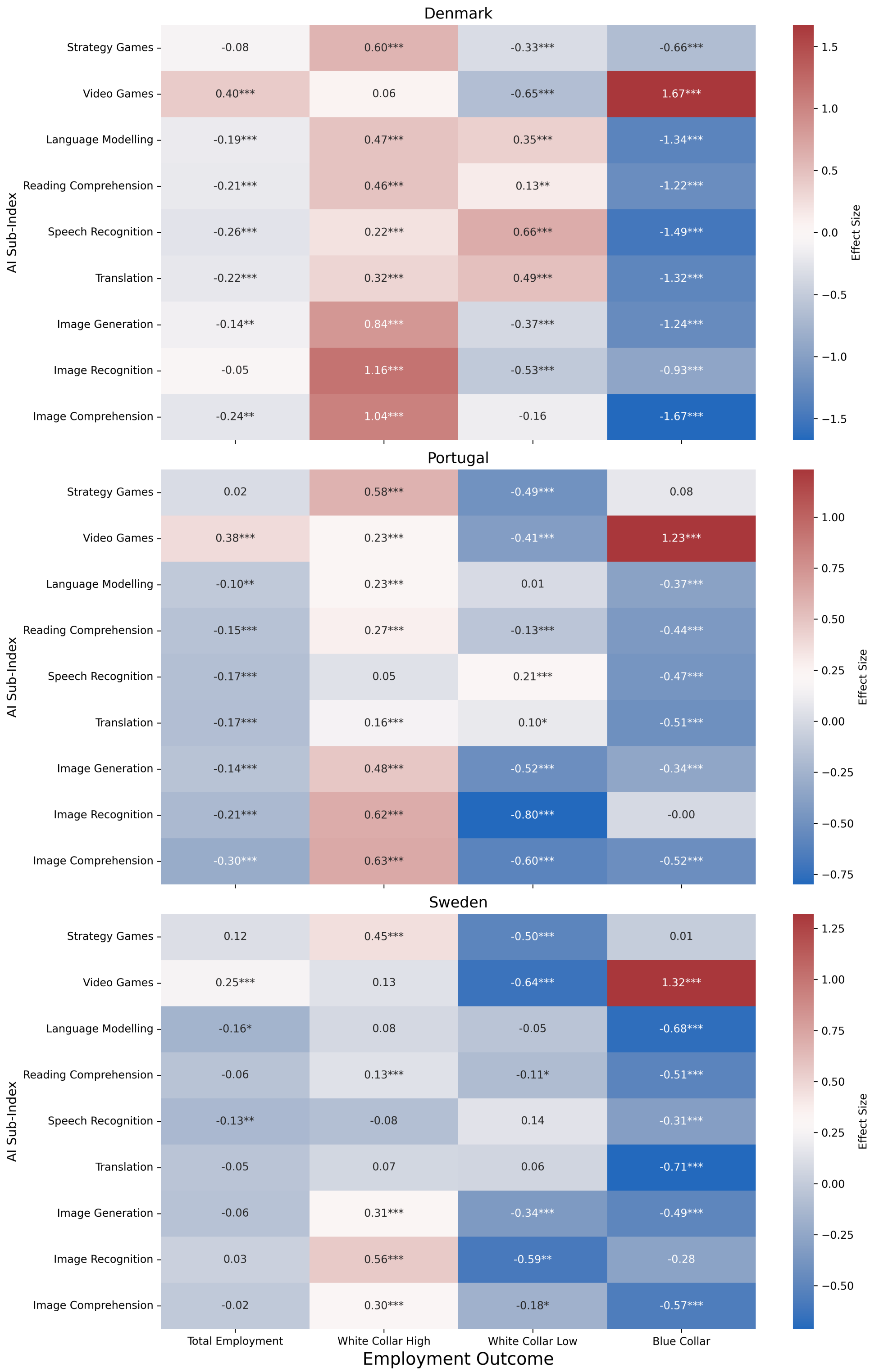


Figure 8: Estimated Associations Between AI Subdomain Exposure and Employment Outcomes

Notes: The heatmap shows estimated coefficients from regressions of log employment on AI subdomain exposure by occupational group. Exposure is based on firm-specific pre-sample 4-digit ISCO08 shares. All models include 3-digit NACE industry-year and location-year fixed effects. Regressors are lagged $(t-1)$, except for firm age. All continuous variables are in logs. Asterisks denote significance (*p $<$ 0.10, **p $<$ 0.05, ***p $<$ 0.01).

subdomains, but tend to be negative overall, in line with the aggregate patterns reported in Figure 7. Subdomains such as strategy games and video games are negatively associated with employment in this group across all three countries, likely reflecting the automation of relatively routine cognitive tasks. By contrast, language-related technologies reveal positive associations in Denmark, particularly for subdomains such as language modelling and translation. These are largely absent in Portugal and Sweden. One interpretation is that such technologies complement certain linguistic tasks that remain labour-intensive in specific industries or public-sector settings. This cross-country variation highlights that even within the same occupational group, the impact of AI exposure is far from uniform.

Substantial heterogeneity is also observed among blue-collar occupations. The estimated associations range from strongly negative—for subdomains such as image recognition, image compression, and speech recognition—to clearly positive in the case of video game technologies. The negative patterns are suggestive of automation in routine manual tasks, while the positive associations may reflect labour augmentation via remote-controlled systems or AI-assisted machinery. Interestingly, the magnitude of the positive associations for video games is strongest for blue-collar workers, exceeding that observed in other occupational groups. This underscores the dual role of AI technologies in this segment: posing automation risks, but also opening up new forms of technologically assisted work.

Taken together, the results underline the importance of unpacking the nature of AI exposure across both occupational categories and technological subdomains. While the overall employment effect for blue-collar workers is statistically insignificant, our disaggregated analysis suggests this average masks opposing associations across AI technologies. We also find that AI is less strongly associated with labour demand in Sweden, relative to Denmark and Portugal. The comparative scope of our data—covering three countries with distinct labour market institutions and industry profiles—allows us to uncover these nuanced patterns, which may not be visible in single-country or more aggregated studies.

5.3. *Implications for the Skill Ratio*

The preceding analysis reveals a nuanced landscape regarding AI's association with employment across occupational groups. While firms tend to expand employment in white-collar-high occupations as AI exposure increases, the evidence is more mixed for white-collar-low and blue-collar roles. The subdomain-level results further suggest that different AI technologies are linked to both complementary and substitutive dynamics across job types. These patterns prompt a pertinent inquiry: how does AI exposure relate to shifts in the overall skill composition of the workforce?

To explore this, we estimate Equation 12 using the firm-level high-to-low skill employment ratio as the dependent variable.[37] High-skilled workers are defined as those with at least a tertiary education degree. The evolution of this skill ratio provides insight into whether AI technologies are associated with a shift in the composition of the workforce toward more complex, knowledge-intensive roles, consistent with theories of skill-biased technological change (Acemoglu and Autor, 2011).

Figure 9 summarises the results. Across all three countries, there is a clear and statistically significant positive association between AI exposure and the skill ratio, indicating that firms exposed to advancing AI capabilities tend to increase the relative share of high-skilled workers. Full regression results are reported in Table A7 in the Appendix.

Among the three countries, Portugal exhibits the strongest association, while being somewhat less precise. The magnitude of the estimated coefficient suggests a relatively steep adjustment toward upskilling. This may reflect the nature of AI adoption in Portuguese firms, where AI applications could be substituting for less complex tasks, thereby increasing the demand for workers capable of interacting with or supervising such technologies. Portugal's comparatively lower baseline of educational attainment and digital intensity (see Online Appendix Figures J1–J2) may also imply more room for such compositional adjust-

[37] The only change from the main specification is the exclusion of the control for the share of workers with tertiary education.

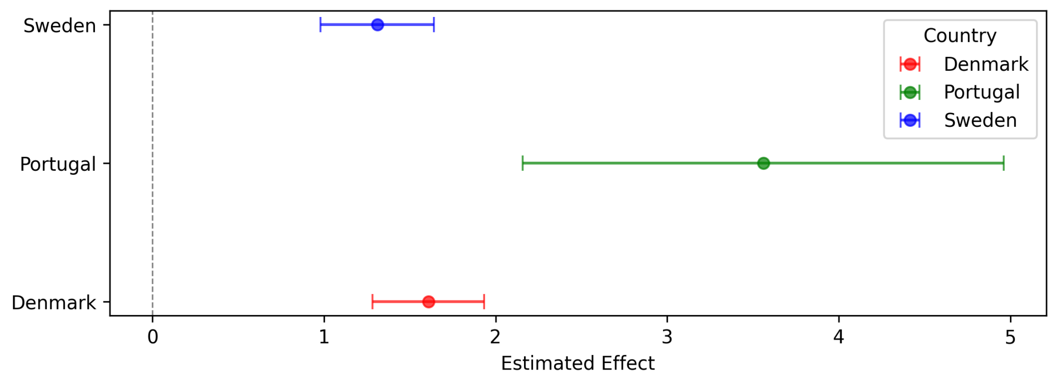


Figure 9: Estimated Associations Between AI Exposure and the Skill Ratio

Notes: The whisker plot shows estimated coefficients and 95% confidence intervals from regressions of the high-to-low skill ratio on AI exposure across Denmark, Portugal, and Sweden. Exposure is measured using the DAIOE index, fixed at firm-specific baseline occupational shares. Models include 3-digit industry-year and location-year fixed effects. Regressors are lagged by one period ($t-1$), except for firm age.

ments.

Denmark and Sweden display smaller but still significant associations. These more moderate shifts may reflect the fact that both countries already operate with a more skill-intensive workforce and higher baseline levels of productivity and digital integration. In such settings, the marginal impact of AI on workforce composition may be more incremental, perhaps reflecting continued but slower upskilling within already complex organisational structures.

In a nutshell, these findings suggest that AI exposure is associated with a systematic shift in firm-level employment structures toward higher-skilled labour.[38] The trend we observe appears robust across distinct institutional contexts but is likely moderated by country-specific factors such as existing skill composition, sectoral structure, and the maturity of digital infrastructures.

[38] Long-difference estimates yield similar up-skilling effects—albeit smaller in magnitude—confirming that our core findings withstand alternative specifications (Table A8). Our findings also echo recent U.S. evidence showing a shift toward general skill upgrading in the labour market (Deming *et al.*, 2025).

6. Concluding Remarks

We develop a new, Dynamic AI Occupational Exposure (DAIOE) measure—tracking annual advances across nine AI subdomains from 2010 to 2023—and link it to rich employer–employee data in Denmark, Portugal, and Sweden.

Our main finding is that firms with higher AI exposure show no systematic change in overall headcounts, but do shift their workforce mix toward a more skilled workforce. In particular, greater DAIOE scores are consistently associated with higher ratios of high-to-low-skill employees, even after accounting for robots, software, and offshoring risks.

Disaggregated results reveal that AI exposure correlates positively with high-skill white-collar employment and negatively with low-skill clerical positions. Blue-collar effects average out to near zero but mask striking variation: "video-game" AI progress tends to raise blue-collar shares, whereas vision and language advances often coincide with reductions.

These patterns underscore why it matters to unpack "AI" into its component technologies. As AI continues to evolve—especially with generative models—DAIOE offers a straightforward way to anticipate both broad up-skilling trends and more focused displacement risks. Policymakers and firms should therefore track developments by subdomain and target training, innovation, and adjustment policies to the uneven terrain of AI exposure across occupations and countries.

Future work could apply DAIOE to other national settings, examine longer-run effects on wages and career paths, or link exposure to firms' investment and innovation outcomes, further deepening our understanding of AI's role in today's labour markets.

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Appendix

A Descriptive Statistics

Table A1

Workforce Composition by Occupational Group

ISCO 1 dig.	Occupation	Denmark	Portugal	Sweden
1	Managers	0.059	0.048	0.072
2	Professionals	0.143	0.084	0.147
3	Technicians and Assoc. Profes.	0.134	0.108	0.152
4	Clerical support workers	0.097	0.136	0.099
5	Services and Sales Workers	0.181	0.186	0.183
6	Skilled Agricultural Forestry and Fishery workers	0.003	0.014	0.006
7	Craft and related trade workers	0.144	0.205	0.135
8	Plant and Machine Operators and Assemblers	0.100	0.103	0.125
9	Elementary Occup.	0.138	0.116	0.080

Notes: The table reports employment shares by 1-digit ISCO-08 occupational code for the estimation samples of the three countries.

Table A2

Workforce Composition by Industry

Industry	Denmark	Portugal	Sweden
Agriculture	0.000	0.025	0.009
Mining	0.003	0.004	0.003
Manufacturing	0.249	0.225	0.230
Utility services	0.000	0.006	0.020
Construction	0.103	0.121	0.100
Trade	0.334	0.306	0.252
Hotel, restaurants	0.039	0.108	0.037
Information and communication	0.075	0.020	0.068
Real estate	0.011	0.013	0.019
Other business services	0.170	0.097	0.155
Public administration	0.000	0.053	0.101
Other services	0.002	0.023	0.006

Notes: The table reports employment shares by industries for the estimation samples of the three countries. Most public-sector firms for Denmark are excluded from the estimation sample due to missing data on capital.

Table A3

Initial and final occupational composition by firm's initial AI exposure

	Denmark				Portugal				Sweden			
	Low exp		High exp		Low exp		High exp		Low exp		High exp	
ISCO 1-digit	2011	2020	2011	2020	2011	2020	2011	2020	2011	2020	2011	2020
1	4.5	4.7	4.2	5.0	3.8	3.8	5.6	5.2	4.1	4	9.2	10.7
2	1.6	2.3	15.1	16.7	1.4	2.2	9.6	9.7	1.3	3.1	33.5	39.5
3	4.7	4.8	12.1	12.6	4.1	4.9	10.4	10.3	2.9	3.7	30.8	24.9
4	3.9	5.8	10.1	11.0	3.7	6.0	11.4	11.6	3.7	2.3	14.9	14.7
5	32.2	33.4	22.7	19.7	26.7	25.8	21.9	21.7	26.8	43.8	2.5	2.3
6	0.5	0.6	0.4	0.4	2.6	2.3	1.9	1.8	1.9	1.5	0.1	0.1
7	19.0	17.8	14.1	13.6	29.4	28.0	18.4	18.3	17.4	14	2.8	3.2
8	8.8	8.8	6.1	6.0	10.8	10.3	6.9	7.0	18.6	9.7	4.8	3.9
9	24.8	21.8	15.3	14.9	17.4	16.8	13.9	14.4	23.3	18.1	1.4	0.8
Avg. firm size	27.9	32.6	12.8	16.6	20.5	22.3	9.5	11.8	20.7	26.6	39.9	47.7

Notes: Percentage of employment within each 1-digit ISCO occupation, by country, firm AI exposure, and year. Firm AI exposure is measured as the employment-weighted DAIOE score. Firms with low AI exposure were in the lowest quartile of AI exposure in 2011, while firms with high AI exposure were in the highest quartile. Firm size is measured by the number of employees.

B Main Results – Tables

Table A4

AI Exposure and Log Employment

	Denmark	Portugal	Sweden
DAIOE	0.383	0.125	0.013
	(0.236)	(0.086)	(0.177)
Robot	0.129***	0.082***	0.185***
	(0.024)	(0.017)	(0.019)
Software	-0.062***	-0.074***	-0.061***
	(0.012)	(0.020)	(0.022)
Offshorability	-0.068***	-0.386***	-0.217
	(0.014)	(0.146)	(0.225)
Sales	0.696***	0.721***	0.552***
	(0.040)	(0.023)	(0.038)
Share Bachelor	-0.195**	-0.654***	0.210**
	(0.079)	(0.095)	(0.087)
Physical Capital Intensity	-0.017	-0.055***	-0.031***
	(0.013)	(0.006)	(0.006)
Firm Age	0.002**	0.004***	0.003**
	(0.001)	(0.000)	(0.001)
Foreign-owned	0.050	0.120***	0.065**
	(0.035)	(0.046)	(0.029)
Exports	0.001	-0.002	0.032***
	(0.002)	(0.004)	(0.005)
Imports	0.036***	0.008	0.039***
	(0.007)	(0.005)	(0.006)
Observations	303,232	773,957	565,736
R-squared	0.906	0.856	0.882
Within R2	0.803	0.778	0.761
No. of firms	51,230	126,231	95,510

Notes: This table reports estimated associations between firm-level AI exposure and the log of total employment. The DAIOE measure is the standardised and weighted average AI exposure of the firm, calculated using firm-specific 4-digit ISCO08 occupational shares from the first year of observation. The Robot and Software exposure measures are constructed analogously, based on occupational exposure scores from Webb (2019); the Offshorability measure draws on occupational exposure from Autor and Dorn (2013). All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels. All regressors are lagged by one period ($t-1$), except for contemporaneous firm age. Continuous variables are in logarithmic form. Regressions are weighted by firm employment in the baseline year. Standard errors, clustered at the firm level, are reported in parentheses. *, **, and *** indicate significance at the 10%, 5%, and 1% levels, respectively.

Table A5

AI Exposure and Log Employment by Occupational Group

	Denmark			Portugal			Sweden		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Dep. var.	Blue collar	White-C. low	White-C. high	Blue collar	White-C. low	White-C. high	Blue collar	White-C. low	White-C. high
DAIOE	0.345	-0.4838	0.8632***	0.559***	-0.668***	0.822***	0.321	-0.735***	0.363**
	(0.272)	(0.310)	(0.268)	(0.151)	(0.102)	(0.103)	(0.237)	(0.263)	(0.166)
Robot	0.615***	-0.100***	-0.100***	0.413***	-0.153***	-0.156***	0.549***	0.056*	-0.081***
	(0.079)	(0.034)	(0.029)	(0.035)	(0.023)	(0.024)	(0.038)	(0.033)	(0.026)
Software	-0.205***	-0.102***	0.037**	-0.094*	-0.135***	0.066***	-0.057	-0.118***	0.017
	(0.061)	(0.021)	(0.017)	(0.052)	(0.026)	(0.017)	(0.041)	(0.034)	(0.028)
Offshorability	-0.231***	0.136***	0.005	-2.572***	2.187***	-0.845***	-4.245***	2.308***	-0.277
	(0.034)	(0.023)	(0.020)	(0.259)	(0.193)	(0.187)	(0.627)	(0.548)	(0.316)
Sales	0.623***	0.631***	0.645***	0.582***	0.660***	0.598***	0.411***	0.485***	0.495***
	(0.035)	(0.035)	(0.038)	(0.051)	(0.026)	(0.019)	(0.028)	(0.030)	(0.033)
Share Bachelor	-1.549***	-1.365***	1.0792***	-1.916***	-1.809***	1.233***	-2.067***	-1.651***	1.124***
	(0.135)	(0.104)	(0.095)	(0.169)	(0.094)	(0.088)	(0.191)	(0.140)	(0.125)
Physical Capital	-0.021	-0.005	-0.000	-0.033***	-0.032***	-0.028***	-0.035***	-0.024***	-0.005
	(0.014)	(0.015)	(0.014)	(0.010)	(0.008)	(0.008)	(0.007)	(0.007)	(0.006)
Firm Age	0.004**	0.003**	0.002*	0.006***	0.005***	0.005***	-0.003	-0.001	0.001
	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.002)	(0.002)	(0.002)
Foreign Owned	-0.154**	-0.002	0.064	-0.141	-0.071	0.215***	0.171**	0.154***	0.132***
	(0.068)	(0.044)	(0.044)	(0.121)	(0.063)	(0.064)	(0.066)	(0.057)	(0.036)
Exports	0.004	-0.003	0.008***	0.008	-0.013***	0.005	0.031***	0.021***	0.038***
	(0.004)	(0.003)	(0.003)	(0.007)	(0.004)	(0.005)	(0.005)	(0.005)	(0.005)
Imports	0.024***	0.039***	0.037***	-0.008	0.015***	0.009**	0.031***	0.036***	0.049***
	(0.007)	(0.007)	(0.007)	(0.013)	(0.006)	(0.004)	(0.005)	(0.005)	(0.006)
Obs.	303,232	303,232	303,232	773,957	773,957	773,957	565,736	565,736	565,736
R^2	0.786	0.840	0.873	0.717	0.819	0.816	0.809	0.868	0.376
Within R^2	0.532	0.622	0.709	0.678	0.681	0.709	0.534	0.584	0.710
Firms	51,230	51,230	51,230	126,231	126,231	126,231	95,510	95,510	95,510

Notes: This table reports estimated associations between firm-level AI exposure and the log of employment, disaggregated by occupational group. "Blue collar" occupations correspond to ISCO-08 major groups 6, 7, 8, and 9; "white collar low" includes groups 4 and 5; and "white collar high" includes groups 1, 2, and 3. The DAIOE measure is the standardised, weighted average AI exposure of the firm, based on 4-digit ISCO-08 occupational shares from the first year of observation. All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels. Regressors are lagged one period ($t-1$), except for contemporaneous firm age. Continuous variables are in logarithmic form. Regressions are weighted by firm employment in the baseline year. Standard errors, clustered at the firm level, are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A6

AI Exposure to Specific Technologies and Log Employment

	Denmark				Portugal				Sweden			
Dep. var.	(1) Total empl.	(2) Blue Collar	(3) White-C. low	(4) White-C. high	(1) Total empl.	(2) Blue Collar	(3) White-C. low	(4) White-C. high	(1) Total empl.	(2) Blue Collar	(3) White-C. low	(4) White-C. high
stratgames	-0.083	-0.660***	-0.331***	0.600***	0.016	0.082	-0.493***	0.584***	0.120	0.010	-0.499***	0.449***
	(0.060)	(0.101)	(0.088)	(0.077)	(0.044)	(0.103)	(0.056)	(0.059)	(0.098)	(0.144)	(0.152)	(0.094)
videogames	0.399***	1.673***	-0.649***	0.061	0.376***	1.233***	-0.408***	0.232***	0.255***	1.321***	-0.640***	0.134
	(0.127)	(0.158)	(0.175)	(0.162)	(0.058)	(0.132)	(0.075)	(0.073)	(0.096)	(0.158)	(0.148)	(0.116)
imgrec	-0.051	-0.932***	-0.529***	1.158***	-0.210***	-0.001	-0.800***	0.617***	0.028	-0.275	-0.592**	0.557***
	(0.126)	(0.189)	(0.170)	(0.144)	(0.072)	(0.163)	(0.091)	(0.104)	(0.111)	(0.212)	(0.234)	(0.115)
imgcompr	-0.240**	-1.673***	-0.159	1.042***	-0.300***	-0.521***	-0.601***	0.631***	-0.021	-0.567***	-0.176*	0.300***
	(0.114)	(0.158)	(0.140)	(0.130)	(0.075)	(0.158)	(0.098)	(0.102)	(0.055)	(0.102)	(0.101)	(0.067)
imggen	-0.141**	-1.244***	-0.367***	0.839***	-0.144***	-0.342***	-0.515***	0.480***	-0.063	-0.495***	-0.343***	0.306***
	(0.059)	(0.116)	(0.077)	(0.074)	(0.053)	(0.112)	(0.063)	(0.075)	(0.068)	(0.108)	(0.108)	(0.084)
readcompr	-0.206***	-1.219***	0.135**	0.459***	-0.154***	-0.441***	-0.132***	0.274***	-0.056	-0.510***	-0.106*	0.132***
	(0.049)	(0.066)	(0.058)	(0.061)	(0.036)	(0.069)	(0.051)	(0.045)	(0.038)	(0.075)	(0.059)	(0.047)
lngmod	-0.194***	-1.338***	0.354***	0.467***	-0.103**	-0.371***	0.005	0.227***	-0.158*	-0.680***	-0.047	0.077
	(0.075)	(0.087)	(0.082)	(0.089)	(0.046)	(0.078)	(0.063)	(0.047)	(0.087)	(0.149)	(0.126)	(0.092)
translat	-0.217***	-1.316***	0.489***	0.318***	-0.174***	-0.513***	0.097*	0.156***	-0.048	-0.712***	0.063	0.068
	(0.059)	(0.073)	(0.065)	(0.074)	(0.045)	(0.076)	(0.059)	(0.055)	(0.043)	(0.098)	(0.094)	(0.055)
speechrec	-0.259***	-1.493***	0.659***	0.218***	-0.172***	-0.468***	0.212***	0.050	-0.130**	-0.311***	0.141	-0.077
	(0.054)	(0.077)	(0.064)	(0.074)	(0.050)	(0.107)	(0.058)	(0.078)	(0.062)	(0.101)	(0.089)	(0.061)

Notes: Each cell reports a coefficient from a separate regression. The nine DAIOE measures represent the firm's standardised and weighted average AI exposure across different subdomains, calculated using 4-digit ISCO-08 occupational shares from the first year of observation. All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels and are weighted by firm employment in the baseline year. Regressors are lagged by one period ($t-1$), except for contemporaneous firm age. Continuous variables are expressed in logarithmic form. Standard errors, clustered at the firm level, are reported in parentheses. *, **, and *** indicate statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A7

AI Exposure and the High-to-Low Skill Ratio

	High-Low Skill Ratio		
	Denmark	Portugal	Sweden
DAIOE	1.609***	3.560***	1.310***
	(0.166)	(0.715)	(0.169)
Robot	-0.154***	-0.074	-0.228***
	(0.019)	(0.104)	(0.019)
Software	0.035**	-0.001	0.018
	(0.015)	(0.061)	(0.017)
Offshorability	-0.016	1.974**	0.369
	(0.021)	(0.926)	(0.307)
Sales	-0.060***	0.052*	-0.015
	(0.015)	(0.031)	(0.009)
Physical Capital	-0.011	-0.048***	0.034***
	(0.008)	(0.018)	(0.006)
Firm Age	-0.002**	-0.003*	-0.001
	(0.001)	(0.001)	(0.001)
Foreign-owned	0.024	-0.219	-0.096***
	(0.042)	(0.336)	(0.033)
Exports	0.009***	0.055***	0.000
	(0.002)	(0.019)	(0.003)
Imports	-0.000	0.004	-0.002
	(0.003)	(0.015)	(0.002)
Obs.	300,277	758,692	558,286
R^2	0.461	0.066	0.376
Within R^2	0.029	0.006	0.029
Firms	50,830	124,316	94,582

Notes: This table reports estimated associations between firm-level AI exposure and the ratio of high-skilled to low-skilled workers within firms. High-skilled workers are defined as those with at least a tertiary education degree. The DAIOE measure is the standardized and weighted average AI exposure of each firm, based on 4-digit ISCO-08 occupational shares from the first year of observation. All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels and are weighted by firm employment in the baseline year. Regressors are lagged by one period ($t-1$), except for contemporaneous firm age. Continuous variables are expressed in logarithmic form. Standard errors, clustered at the firm level, are shown in parentheses. (The low R^2 for Portugal is due to missing educational information for some observations, while mechanically preserving zeros would result in an R^2 comparable to Denmark and Sweden.) *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

C Robustness checks

Table A8

Long Difference: AI Exposure and Employment Outcomes

	Denmark	Portugal	Sweden
Total Employment	0.060***	0.009	-0.048***
	(0.014)	(0.023)	(0.007)
High-Low Skill Ratio	0.184***	0.244**	0.1189***
	(0.024)	(0.102)	(0.0180)
Blue-Collar Employment	0.096***	0.086***	-0.5340***
	(0.020)	(0.026)	(0.014)
Low-Skill White-Collar Emp.	-0.021	-0.136***	0.1489***
	(0.020)	(0.038)	(0.0150)
High-Skill White-Collar Emp.	0.007	-0.037	0.3017***
	(0.017)	(0.029)	(0.0122)

Notes: Each cell reports a coefficient from a separate regression. The table reports firm-level results from estimating long-difference regressions of Equation 13, where the outcome is the change in log employment (or, for the skill ratio, its DHS-transformed change á la Davis and Haltiwanger (1992)) between 2010–11 and 2020–21, and the AI exposure is the corresponding change in firm-level DAIOE based on initial occupational shares, and including baseline controls and industry-year and location-year fixed effects measured in 2010–11. The DAIOE measure is the standardised and weighted average AI exposure of each firm, based on 4-digit ISCO-08 occupational shares from the first year of observation. All regressions are weighted by firm employment in the baseline year. The number of firms and observations are as follows: for Denmark 17,576; Portugal 50,364; and for Sweden 36,235. Standard errors, clustered at the firm level, are shown in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively.

Table A9

AI Exposure and Log Employment

(Unweighted Specification)

	Denmark	Portugal	Sweden
DAIOE	-0.050**	-0.018	0.208***
	(0.021)	(0.016)	(0.024)
Robot	0.044***	0.022***	0.074***
	(0.004)	(0.003)	(0.004)
Software	-0.012***	0.004	-0.035***
	(0.003)	(0.003)	(0.004)
Offshorability	-0.013***	-0.198***	-0.421***
	(0.004)	(0.027)	(0.037)
Sales	0.603***	0.408***	0.400***
	(0.011)	(0.006)	(0.009)
Share Bachelor	-0.024*	-0.012	0.209***
	(0.014)	(0.012)	(0.017)
Physical Capital Intensity	-0.008***	-0.004***	-0.016***
	(0.002)	(0.001)	(0.001)
Firm Age	0.003***	0.006***	0.002***
	(0.000)	(0.000)	(0.000)
Foreign-owned	0.056***	0.344***	0.206***
	(0.013)	(0.016)	(0.013)
Exports	0.000	0.003***	0.018***
	(0.001)	(0.001)	(0.001)
Imports	0.011***	0.017***	0.018***
	(0.001)	(0.001)	(0.001)
Observations	303,243	773,957	565,736
R-squared	0.702	0.524	0.582
Within R2	0.656	0.469	0.516
No. of firms	51,233	126,231	95,510

Notes: As a robustness check, this table reports estimated associations between firm-level AI exposure and the log of total employment, using unweighted regressions. The DAIOE measure is the standardised, weighted average AI exposure of each firm, based on 4-digit ISCO-08 occupational shares from the first year of observation. All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels. Regressors are lagged by one period ($t-1$), except for contemporaneous firm age. Continuous variables are expressed in logarithmic form. Standard errors, clustered at the firm level, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.

Table A10

AI Exposure and Log Employment by Occupational Group

(Unweighted Specification)

	Denmark			Portugal			Sweden		
	(1)	(2)	(3)	(1)	(2)	(3)	(1)	(2)	(3)
Dep. var.	Blue collar	White-C. low	White-C. high	Blue collar	White-C. low	White-C. high	Blue collar	White-C. low	White-C. high
DAIOE	-0.088***	-0.516***	0.302***	0.147***	-0.445***	0.368***	0.157***	-0.234***	0.413***
	(0.033)	(0.032)	(0.029)	(0.018)	(0.017)	(0.016)	(0.027)	(0.027)	(0.026)
Robot	0.254***	-0.067***	-0.114***	0.272***	-0.137***	-0.119***	0.245***	0.041***	-0.116***
	(0.008)	(0.006)	(0.005)	(0.004)	(0.003)	(0.003)	(0.006)	(0.005)	(0.004)
Software	-0.049***	-0.049***	0.059***	-0.006*	-0.031***	0.063***	-0.017***	-0.105***	0.018***
	(0.007)	(0.005)	(0.005)	(0.004)	(0.003)	(0.003)	(0.005)	(0.004)	(0.004)
Offshorability	-0.159***	0.120***	0.054***	-1.406***	1.397***	-0.379***	-2.519***	1.136***	0.836***
	(0.006)	(0.006)	(0.005)	(0.033)	(0.030)	(0.027)	(0.050)	(0.044)	(0.040)
Sales	0.492***	0.426***	0.520***	0.260***	0.292***	0.262***	0.262***	0.294***	0.317***
	(0.010)	(0.009)	(0.010)	(0.004)	(0.004)	(0.004)	(0.006)	(0.007)	(0.007)
Share Bachelor	-0.645***	-0.663***	0.823***	-0.629***	-0.729***	1.241***	-0.777***	-0.563***	1.025***
	(0.021)	(0.021)	(0.018)	(0.011)	(0.013)	(0.012)	(0.016)	(0.017)	(0.017)
Physical Capital	-0.005***	0.002	0.003*	0.001	0.003***	0.002**	-0.015***	-0.006***	0.001
	(0.002)	(0.002)	(0.002)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Firm Age	0.006***	0.003***	0.005***	0.006***	0.007***	0.006***	0.002***	0.002***	0.002***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Foreign-owned	-0.209***	0.054***	0.108***	-0.011	0.208***	0.405***	0.013	0.088***	0.252***
	(0.019)	(0.017)	(0.015)	(0.019)	(0.018)	(0.016)	(0.014)	(0.014)	(0.013)
Exports	0.000	-0.000	0.007***	0.011***	-0.006***	0.005***	0.017***	0.014***	0.024***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Imports	0.004***	0.015***	0.011***	0.009***	0.022***	0.015***	0.011***	0.017***	0.025***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Obs.	303,243	303,243	303,243	773,957	773,957	773,957	565,736	565,736	565,736
R^2	0.596	0.530	0.678	0.572	0.488	0.551	0.612	0.531	0.658
Within R^2	0.349	0.318	0.506	0.306	0.342	0.418	0.293	0.286	0.442
Firms	51,233	51,233	51,233	126,231	126,231	126,231	95,510	95,510	95,510

Notes: As a robustness check, this table reports estimated associations between firm-level AI exposure and the log of employment, disaggregated by occupational group. "Blue collar" includes ISCO-08 major groups 6, 7, 8, and 9; "white collar low" includes groups 4 and 5; and "white collar high" includes groups 1, 2, and 3. The DAIOE measure is the standardised, weighted average AI exposure of each firm, based on 4-digit ISCO-08 occupational shares from the first year of observation. All regressions include fixed effects at the 3-digit NACE industry-year and location-year levels. No regression weights are applied. Continuous variables are expressed in logarithmic form. Standard errors, clustered at the firm level, are shown in parentheses. *, **, and *** denote statistical significance at the 10%, 5%, and 1% levels, respectively.